White spot syndrome virus Nimaviridae Arthropoda extended

AB855742.1

NCBI FASTA

Genome And Source

LENGTH	TYPE	GC%
528	dsDNA	49.6%
COMPLETENESS	**COUNTRY**	**YEAR**
partial_sequence	Iran	2013

Curation Status

1188 Isolates	41 Hosts
1 Proteins	OK Sequence

Outbreak Events

China (1992)
Global pandemic since 1992; major outbreaks in China, Thailand, India, Ecuador, Mexico; estimated annual losses >$1B

None
Same as WSSV - global pandemic

Reference

Activation of Host Endogenous Reverse Transcriptase in Response to White Spot Syndrome Virus Infection in Penaeus monodon
Wongkhaluang P; Taengchaiyaphum S; Wongpim T; Sukthaworn S; Nakayama H; Thitamadee S; Sritunyalucksana K
Fish & shellfish immunology 2026
PMID: 42055174 DOI: 10.1016/j.fsi.2026.111380

ICTV Mapping

No reviewed species-level mapping is available. Family-level ICTV references:

- Whispovirus lacteolymphae (Whispovirus)
- Whispovirus xiabaidian (Whispovirus)

Reviewed Evidence

Only manual-checked evidence records are displayed here. Candidate or inferred records are excluded from this page.

Virulence / pathogenicity

Auto-extracted: mortality in 'Activation of Host Endogenous Reverse Transcriptase in Response to White Spot Syndrome Virus Infection in Penaeus monodon'
No value text
mortality - medium

Auto-extracted: mortality in 'Decapod iridescent virus 1 (DIV1) enters hematopoietic Cherax quadricarinatus cells via caveola-mediated endocytosis in a pH-dependent manner'
No value text
mortality - medium

Auto-extracted: pathogenicity in 'New genotypes of white spot syndrome virus (WSSV) and Taura syndrome virus (TSV) from the Kingdom of Saudi Arabia.'
No value text
pathogenicity - medium

Auto-extracted: mortality in 'Development and application of a quantitative real-time PCR method for detection of Decapod iridescent virus 1.'
No value text
mortality - medium

Auto-extracted: mortality in 'The first detection of white spot syndrome virus in naturally infected cultured Chinese mitten crabs, Eriocheir sinensis'

No value text
mortality - medium

Auto-extracted: mortality in 'Development of TaqMan qPCR and MIRA detection methods of Hepatospora eriocheir in Chinese mitten crab.'
No value text
mortality - medium

Auto-extracted: mortality in 'Bursicon homodimers enhance the expression of innate immune genes and delay WSSV-induced mortality in Litopenaeusvannamei.'
No value text
mortality - medium

Auto-extracted: mortality in 'Estimation of white spot syndrome virus severity grades dynamics in whiteleg shrimp (Penaeus vannamei) during the disease transition process.'
No value text
mortality - medium

Auto-extracted: mortality in 'Combating White Spot Syndrome Virus (WSSV) in Global Shrimp Farming: Unraveling Its Biology, Pathology, and Control Strategies.'
No value text
mortality - medium

Auto-extracted: mortality in 'Simultaneous detection of five shrimp pathogens using a single-tube EvaGreen real-time PCR assay with differential melting temperature.'
No value text
mortality - medium

Temperature

Auto-extracted: temperature in 'Development of TaqMan qPCR and MIRA detection methods of Hepatospora eriocheir in Chinese mitten crab.'
No value text
temperature - medium

Temperature data found: temperature, thermal, °C
No value text
temperature - medium

The future of cell biology: Emerging model organisms Articulating the “stem cell niche” paradigm through the lens of non-model aquatic invertebrates Physiological basis of interactive responses to temperature and salinity in coastal marine invertebra
No value text
temperature - medium

Prior to experimental challenges, naïve Pacific whiteleg shrimp Penaeus vannamei (2–5g fresh weight) were purchased from local SPF shrimp broodstock producers and acclimated in the laboratory for 2 days in 20 L artificial seawater (Marinium) at 20 pp
No value text
temperature - medium

Defense parameters Long-term trends in oyster population dynamics in Delaware Bay: regime shifts and response to disease Susceptibility of testicular cell cultures of crab, Scylla serrata (Forskal) to white spot syndrome virus Research priorities for
No value text
temperature - medium

Impact of pig slurry treatment on reduction of the viral load and prevalence of the virus in food substrates High prevalence of hepatitis E virus infection in goats Two-year systematic study to assess norovirus contamination in oysters from commercia
No value text
temperature - medium

Meal as Feed Additive for Pacific White Shrimp Reared under Low Temperature and Challenged by WSSV in Association with Thermal Stress.
No value text
temperature - medium

Quantitative Real-Time PCR (RT-qPCR) Analysis Expressions of 10 target immune-related genes, heat shock protein 70 (hsp70), heat shock protein 90 (hsp90), down syndrome cell adhesion molecule (DSCAM), lipopolysaccharide and β-1,3-glucan- binding prot
No value text
temperature - medium

Conventional White Shrimp Challenge Test Before the challenge test, healthy white shrimp (Penaeus vannamei; 4.6 ± 0.48 g) were transferred to 50 L seawater in plastic tanks, and were acclimated for one week under the following conditions: 30 shrimp/t
No value text
temperature - medium

Pertinent examples of pathogens and parasites affecting marine aquaculture and their temperature ranges, optimal conditions, and critical thresholds affecting their environmental survival Disease Pathogen/Parasite (taxonomy) Hosts Temperature range (
No value text
temperature - medium

Hosts

Acetes chinensis
Sergestidae

Alpheus distinguendus
Alpheidae

Astacidea
Multiple

Callinectes arcuatus
Portunidae

Chasmagnathus granulata
Varunidae

Cherax quadricarinatus
Parastacidae

Crangon sp.
Crangonidae

Crustacea
Multiple

Eriocheir sinensis
Varunidae

Fenneropenaeus indicus
Penaeidae

L.vannamei

Penaeidae

Litopenaeus sp.
Penaeidae

Litopenaeus vannamei
Penaeidae

Litopenaeus vannamei
Penaeidae

Litopenaeus vannamei
Penaeidae

Macrobrachium rosenbergii
Palaemonidae

Macrobrachium rosenbergii de Man
Palaemonidae

Marsupenaeus japonicus
Penaeidae

Marsupenaeus japonicus
Penaeidae

Metapenaeopsis lamellata
Penaeidae

Metapenaeus affinis
Penaeidae

Metapenaeus monoceros
Penaeidae

Octolasmis neptuni
Poecilasmatidae

Orisarma dehaani
Sesarmidae

Palaemon gravieri
Palaemonidae

Panulirus echinatus
Palinuridae

Panulirus homarus
Palinuridae

Penaeid shrimp
Penaeidae

Penaeus (Marsupenaeus) japonicus Bate

Penaeus chinensis
Penaeidae

Penaeus indicus
Penaeidae

Penaeus japonicus
Penaeidae

Penaeus merguiensis
Penaeidae

Penaeus monodon
Penaeidae

Penaeus monodon
Penaeidae

Penaeus monodon
Penaeidae

Penaeus setiferus
Penaeidae

Penaeus spp.
Penaeidae

Penaeus spp.
Penaeidae

Penaeus vannamei
Penaeidae

Procambarus alleni
Cambaridae

Procambarus clarkii
Cambaridae

Scylla serrata
Portunidae

Scylla sp.
Portunidae

Trachypenaeus curvirostris
Penaeidae

Trachysalambria curvirostris
Penaeidae

black tiger shrimp
Penaeidae

Diagnostic And Control Records

Diagnostic methods

WSSV-PCR (Lo 1996)
Nucleic acid amplification - VP28/VP26 gene
high

Control methods

Chemical control (trichlorfon) + biosecurity
biosecurity - medium
Trichlorfon validated for controlling white spot disease and enhancing biosecurity protocols

Farm practice management (wild carriers)
biosecurity - medium
Farm practices and wild carriers affect white spot disease in marine shrimp

Feed biosecurity risk assessment (PCR)
biosecurity - medium
PCR assay assesses biosecurity risk of extruded feed for WSSV

PCR-based SPF broodstock screening
biosecurity - high
Preventing vertical transmission through certified SPF broodstock is the most effective WSSV control strategy.

Pathogen risk assessment (WSSV/AHPND)
biosecurity - medium
Assessing WSSV and AHPND biosecurity risk in shrimp

Resilient shrimp aquaculture strategies
biosecurity - medium
Challenges and strategies for globally resilient shrimp aquaculture

SPF broodstock import biosecurity
biosecurity - medium
Dataset on P. vannamei broodstock imports supporting SPF programs

WSSV surveillance sensitivity modelling
biosecurity - medium
Scenario tree model quantifies WSSV surveillance sensitivity

Chlorine dioxide (ClO_2) water disinfection
disinfection - medium
Chlorine dioxide in shrimp-rearing water for pathogen control

Chlorine disinfection (30 ppm)
disinfection - high
30 ppm chlorine kills WSSV-infected shrimp and carriers; water exchange minimized

Beta-glucan
immunostimulant - high
Glucan-induced disease resistance in tiger shrimp (Penaeus monodon)

Beta-glucan / probiotic supplementation
immunostimulant - low
Moderate improvement in survival (10-20% RPS) through non-specific immune enhancement. Variable results across studies.

Plant immunomodulators
immunostimulant - medium
Immunomodulatory plants against WSSV in shrimp culture (review)

Probiotic (Lactobacillus johnsonii KD1)
immunostimulant - medium
Probiotic Lactobacillus johnsonii KD1 against shrimp pathogen

Probiotic strains
immunostimulant - medium
Four probiotic strains improve survival of L. vannamei after challenge

Probiotics/prebiotics/synbiotics
immunostimulant - medium
Probiotics, prebiotics and synbiotics for shrimp aquaculture (review)

Sulfated galactans
immunostimulant - medium
Immunostimulatory sulfated galactans from Gracilaria induce resistance to WSSV

Pond fallowing (WSSV decay)
pond_management - medium
WSSV decay kinetics in seawater/sediment inform reservoir management and fallowing

Pond management SOP integration
pond_management - medium
SOPs harmonize pond management, water quality and disease resilience

Early selection (3-5 days post-infection)
selective_breeding - high
High genetic correlation of WSSV resistance across infection times enables early selection

Genetic selection for WSSV resistance
selective_breeding - high
Genetic variation in disease resistance against WSSV in Litopenaeus vannamei

High water temperature (33 C)
thermal_management - high
33 C water reduces clinical and virological outcome of WSSV infection

Temperature elevation (32-33C)
thermal_management - high
Elevated water temperature suppresses WSSV replication and delays mortality. Not curative but can buy time for harvest.

WSSV vaccination (review)
vaccine - medium
Vaccination studies and innate immune memory in shrimp

WSSV vaccine development
vaccine - medium
Recent progress in WSSV vaccines for protecting shrimp against infection

Proteins

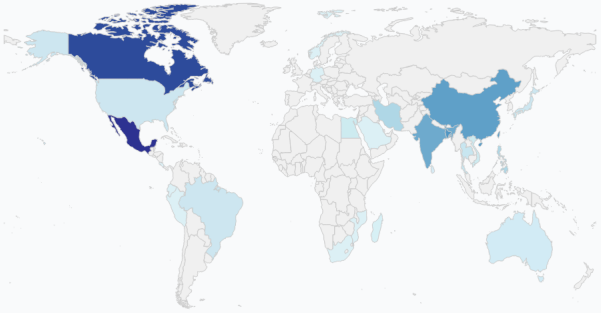
NAME	GENE	LENGTH	FUNCTIONAL CATEGORY	3D STRUCTURE
structural protein VP28	-	175 aa	structural	-

1 proteins total

Related Literature (1860 articles)

Activation of Host Endogenous Reverse Transcriptase in Response to White Spot Syndrome Virus Infection in Penaeus monodon
Wongkhaluang P; Taengchaiyaphum S; Wongpim T; Sukthaworn S; Nakayama H; Thitamadee S; Sritunyalucksana K
Fish & shellfish immunology 2026
PMID: 42055174 DOI
Additional 1 evidence source articles (PMID)

Geographic Distribution



Mexico 221 isolates
Canada 198 isolates
Bangladesh 124 isolates
China 117 isolates
India 109 isolates
Philippines 67 isolates
Iran 67 isolates
Thailand 33 isolates

Taxonomy Information

FAMILY
Nimaviridae

HOST PHYLUM
Arthropoda

GENOME TYPE
dsDNA

DISCOVERY METHOD
isolated_and_cultured

Related Isolates

ACCESSION	LENGTH	GC	COMPLETENESS	COUNTRY	YEAR
AB855742.1	528	49.6	partial_sequence	Iran	2013
AB855742.1	528	49.6	partial_sequence	Iran	N/A
AB974691.1	366	51.1	gene_fragment	Iran	2011
AF099142.1	11,319	42.2	partial_sequence	Canada	2023
AF132668.1	8,715	40.5	gene_fragment	China	2000
AF132669.1	4,710	42.7	gene_fragment	China	2000
AF144620.1	1,242	42.9	gene_fragment	Canada	2023
AF173992.1	910	38.8	gene_fragment	Canada	2023
AF173993.1	1,120	44.0	gene_fragment	Canada	2023
AF178573.1	2,833	38.3	unknown	Canada	2023
AF227910.1	306	43.8	mRNA	Canada	2023

ACCESSION	LENGTH	GC	COMPLETENESS	COUNTRY	YEAR
AF227911.1	1,373	42.5	mRNA	Canada	2023
AF228518.1	1,070	36.9	gene_fragment	Canada	2023
AF267731.1	1,819	42.4	gene_fragment	China	2000
AF272979.1	1,250	42.9	gene_fragment	N/A	2000
AF272980.1	1,450	40.3	gene_fragment	N/A	2000
AF295123.1	1,416	38.4	unknown	Canada	2023
AF295124.1	510	46.5	unknown	Canada	2023
AF308164.1	726	39.9	mRNA	Canada	2023
AF332093.3	305,119	41.0	complete_genome	China	2001
AF335541.1	2,520	41.0	gene_fragment	China	2000
AF343568.1	14,380	40.4	gene_fragment	Canada	2023
AF343569.1	2,252	38.8	unknown	Canada	2023
AF361753.1	1,371	45.4	partial_sequence	Canada	2023
AF365882.1	707	42.7	partial_sequence	Canada	2023
AF369029.2	292,967	41.1	complete_genome	Thailand	2001
AF376806.1	671	34.4	mRNA	N/A	2001
AF376807.1	735	48.0	mRNA	N/A	2001
AF380840.1	627	37.6	gene_fragment	Canada	2023
AF380841.1	615	43.1	gene_fragment	Canada	2023
AF380842.1	615	48.5	gene_fragment	Canada	2023
AF394231.1	12,002	38.9	gene_fragment	Canada	2023
AF394232.1	1,905	39.2	gene_fragment	Canada	2023
AF395544.1	591	50.6	gene_fragment	Canada	2023
AF395545.1	1,401	40.4	gene_fragment	Canada	2023
AF402994.1	219	43.8	partial_sequence	Canada	2023
AF402995.1	1,019	42.7	partial_sequence	Canada	2023
AF402996.1	285	44.6	partial_sequence	Canada	2023
AF402997.1	363	51.8	partial_sequence	Canada	2023
AF402998.1	552	42.0	partial_sequence	Canada	2023
AF402999.1	624	38.0	partial_sequence	Canada	2023
AF403000.1	624	43.1	partial_sequence	Canada	2023
AF403001.1	891	38.8	partial_sequence	Canada	2023
AF403002.1	711	57.5	partial_sequence	Canada	2023
AF403003.1	900	37.9	partial_sequence	Canada	2023
AF403004.1	1,071	54.3	partial_sequence	Canada	2023
AF411464.1	204	49.0	partial_sequence	Canada	2023
AF411634.1	843	36.9	partial_sequence	Canada	2023
AF411635.1	1,152	37.4	partial_sequence	Canada	2023
AF411636.1	876	38.7	partial_sequence	Canada	2023
AF411637.1	3,903	42.1	partial_sequence	Canada	2023
AF440570.1	307,287	41.0	complete_genome	N/A	2000
AF493144.1	2,532	44.1	partial_sequence	Canada	2023
AF493145.1	1,632	40.7	partial_sequence	Canada	2023
AF493146.1	1,521	39.0	partial_sequence	Canada	2023
AF493147.1	1,161	42.1	partial_sequence	Canada	2023
AF493148.1	1,011	35.2	partial_sequence	Canada	2023

ACCESSION	LENGTH	GC	COMPLETENESS	COUNTRY	YEAR
AF493149.1	1,086	41.2	partial_sequence	Canada	2023
AF493150.1	960	41.6	partial_sequence	Canada	2023
AF493151.1	648	41.0	partial_sequence	Canada	2023
AF502435.1	615	48.8	gene_fragment	Canada	2023
AF536176.1	348	44.5	gene_fragment	Thailand	1999
AJ937742.1	311	48.5	partial_sequence	N/A	2016
AJ937859.1	516	50.0	partial_sequence	Mexico	2016
AJ937860.1	363	51.2	partial_sequence	Mexico	2016
AJ937861.1	569	43.9	partial_sequence	Mexico	2016
AP027278.1	299,976	41.0	complete_genome	Japan	1995
AP027279.1	293,923	41.1	complete_genome	Japan	1995
AP027280.1	311,562	40.9	complete_genome	Japan	1995
AP027281.1	301,236	41.0	complete_genome	Japan	2017
AP027282.1	301,054	41.1	complete_genome	Japan	2018
AP027283.1	295,104	41.1	complete_genome	Japan	2023
AP027284.1	298,496	41.0	complete_genome	Japan	2020
AP027285.1	289,287	41.1	complete_genome	Japan	2016
AP027286.1	289,353	41.1	complete_genome	Japan	2017
AP027287.1	289,352	41.1	complete_genome	Japan	2017
AP027288.1	288,252	41.1	complete_genome	Japan	2018
AP027289.1	288,494	41.1	complete_genome	Japan	2020
AP027290.1	288,190	41.1	complete_genome	Japan	2021
AY008843.1	2,704	41.2	gene_fragment	Canada	2023
AY044842.1	1,491	40.2	partial_sequence	Canada	2023
AY044843.1	2,400	40.9	partial_sequence	Canada	2023
AY048541.1	1,533	40.8	partial_sequence	Canada	2023
AY048542.1	1,452	39.3	partial_sequence	Canada	2023
AY048543.1	1,344	40.0	partial_sequence	Canada	2023
AY048544.1	1,845	39.4	partial_sequence	Canada	2023
AY048545.1	2,022	40.9	partial_sequence	Canada	2023
AY048546.1	1,869	40.2	partial_sequence	Canada	2023
AY048547.1	5,052	50.7	partial_sequence	Canada	2023
AY050246.1	246	35.4	gene_fragment	Canada	2023
AY050247.1	204	48.5	gene_fragment	Canada	2023
AY050248.1	963	42.1	gene_fragment	Canada	2023
AY117543.1	725	40.7	mRNA	Canada	2023
AY126666.1	7,960	41.0	gene_fragment	Canada	2023
AY160771.1	366	51.1	gene_fragment	Canada	2023
AY168644.1	615	48.6	gene_fragment	Canada	2023
AY220743.1	243	50.2	gene_fragment	Canada	2023
AY220744.1	366	51.1	gene_fragment	Canada	2023
AY220745.1	627	38.1	gene_fragment	Canada	2023
AY220746.1	615	43.2	gene_fragment	Canada	2023
AY245783.1	411	40.4	mRNA	N/A	2016
AY245784.1	305	43.9	mRNA	N/A	2016
AY245785.1	509	41.6	mRNA	N/A	2016

ACCESSION	LENGTH	GC	COMPLETENESS	COUNTRY	YEAR
AY245786.1	636	46.9	mRNA	N/A	2016
AY245787.1	570	38.8	mRNA	N/A	2016
AY245788.1	387	41.3	mRNA	N/A	2016
AY245789.1	440	38.6	mRNA	N/A	2016
AY245790.1	205	36.6	mRNA	N/A	2016
AY249434.1	715	45.2	mRNA	China	2012
AY249435.1	641	42.6	gene_fragment	China	2012
AY249436.1	641	42.6	gene_fragment	China	2012
AY249437.1	641	42.6	gene_fragment	N/A	2012
AY249438.1	641	42.6	gene_fragment	China	2012
AY249439.1	641	42.6	gene_fragment	Philippines	2012
AY249440.1	643	47.6	gene_fragment	China	2012
AY249441.1	638	48.0	gene_fragment	Philippines	2012
AY249442.1	654	47.4	gene_fragment	N/A	2012
AY249443.1	659	47.3	gene_fragment	China	2012
AY249444.1	408	48.0	gene_fragment	China	2012
AY249445.1	408	48.0	gene_fragment	China	2012
AY249446.1	408	48.0	gene_fragment	N/A	2012
AY249447.1	408	48.0	gene_fragment	China	2012
AY249448.1	408	48.0	gene_fragment	Philippines	2012
AY249449.1	413	40.0	gene_fragment	China	2012
AY249450.1	413	40.0	gene_fragment	China	2012
AY249451.1	413	40.0	gene_fragment	N/A	2012
AY249452.1	413	40.0	gene_fragment	China	2012
AY249453.1	413	40.0	gene_fragment	Philippines	2012
AY249454.1	724	35.5	gene_fragment	China	2012
AY249455.1	723	35.4	gene_fragment	China	2012
AY249456.1	730	35.6	gene_fragment	N/A	2012
AY249457.1	738	35.0	gene_fragment	China	2012
AY249458.1	740	35.1	gene_fragment	Philippines	2012
AY316119.1	366	50.8	gene_fragment	N/A	2004
AY324168.1	422	39.1	partial_sequence	Canada	2023
AY324169.1	429	39.4	partial_sequence	Canada	2023
AY324170.1	424	39.6	partial_sequence	Canada	2023
AY324171.1	414	39.4	partial_sequence	Canada	2023
AY324881.1	615	48.6	gene_fragment	N/A	2004
AY325896.1	687	41.9	gene_fragment	N/A	2004
AY374120.1	186	56.5	gene_fragment	N/A	2004
AY374443.1	505	41.4	partial_sequence	N/A	2003
AY374444.1	492	42.3	partial_sequence	N/A	2003
AY422226.1	186	56.5	gene_fragment	India	2008
AY422227.1	366	50.5	gene_fragment	India	2008
AY422228.1	615	48.3	gene_fragment	India	2008
AY422229.1	627	38.0	gene_fragment	India	2008
AY422230.1	615	43.1	gene_fragment	India	2008
AY517490.1	846	37.0	gene_fragment	Canada	2023

ACCESSION	LENGTH	GC	COMPLETENESS	COUNTRY	YEAR
AY682926.1	615	48.5	gene_fragment	China	2004
AY713371.1	178	57.3	partial_sequence	N/A	2016
AY713372.1	290	50.7	partial_sequence	N/A	2016
AY713373.1	301	39.9	partial_sequence	N/A	2016
AY713374.1	984	41.9	partial_sequence	N/A	2016
AY753327.1	7,800	38.7	complete_genome	Thailand	2005
AY850066.1	896	46.3	partial_sequence	United States	2016
AY864664.1	505	37.2	partial_sequence	Thailand	2005
AY864665.1	688	38.7	partial_sequence	Thailand	2005
AY864666.1	918	36.9	partial_sequence	Thailand	2005
AY864667.1	696	44.7	partial_sequence	Thailand	2005
AY864668.1	1,105	54.5	partial_sequence	Thailand	2005
AY864669.1	682	45.9	partial_sequence	Thailand	2005
AY864670.1	652	47.7	partial_sequence	Thailand	2005
AY870344.1	306	36.6	partial_sequence	Iran	2016
AY873785.1	615	48.6	gene_fragment	India	2005
AY873786.1	366	50.8	gene_fragment	India	2005
AY884234.1	1,102	36.6	gene_fragment	China	2001
AY897233.1	966	40.4	mRNA	China	2001
AY897234.1	3,027	42.3	mRNA	China	2001
AY897235.1	920	35.5	mRNA	China	2001
BD453188.1	366	51.1	gene_fragment	Canada	2023
BD453189.1	186	56.5	gene_fragment	Canada	2023
CK738633.1	255	34.5	mRNA	Canada	2023
CN950623.1	350	32.3	mRNA	Canada	2023
CV067812.1	265	42.3	mRNA	Canada	2023
DI176664.1	733	41.6	unknown	Canada	2023
DI176665.1	24	N/A	unknown	N/A	N/A
DI176666.1	22	N/A	unknown	N/A	N/A
DI176667.1	21	N/A	unknown	N/A	N/A
DI493636.1	21	N/A	unknown	N/A	N/A
DI493637.1	20	N/A	unknown	N/A	N/A
DI493638.1	22	N/A	unknown	N/A	N/A
DI493639.1	20	N/A	unknown	N/A	N/A
DI493640.1	21	N/A	unknown	N/A	N/A
DI493641.1	19	N/A	unknown	N/A	N/A
DI493642.1	34	N/A	unknown	N/A	N/A
DI493643.1	34	N/A	unknown	N/A	N/A
DI493644.1	35	N/A	unknown	N/A	N/A
DI493645.1	35	N/A	unknown	N/A	N/A
DI493646.1	33	N/A	unknown	N/A	N/A
DI493647.1	33	N/A	unknown	N/A	N/A
DN796202.1	608	48.5	mRNA	Canada	2023
DN796673.1	718	46.1	mRNA	Canada	2023
DQ007314.1	368	51.4	gene_fragment	China	2001
DQ007315.1	615	48.6	gene_fragment	China	2001

ACCESSION	LENGTH	GC	COMPLETENESS	COUNTRY	YEAR
DQ013881.1	615	48.6	gene_fragment	India	1997
DQ013882.1	615	48.6	gene_fragment	India	2016
DQ013883.1	615	48.8	gene_fragment	India	2003
DQ098011.1	615	48.6	gene_fragment	China	2005
DQ121373.1	805	38.4	mRNA	Canada	2023
DQ196431.1	715	36.4	gene_fragment	Iran	2016
DQ681068.1	736	35.6	gene_fragment	India	2008
DQ681069.1	615	48.6	gene_fragment	India	2008
DQ681070.1	657	42.8	partial_sequence	India	2008
DQ681071.1	407	48.4	gene_fragment	India	2008
DQ681072.1	386	40.7	gene_fragment	India	2008
DQ902654.1	241	51.0	partial_sequence	India	2008
DQ902655.1	366	50.8	gene_fragment	India	2008
DQ902656.1	627	37.8	gene_fragment	India	2008
DQ902657.1	616	43.2	partial_sequence	India	2008
DQ902658.1	615	48.6	gene_fragment	India	2008
DQ902659.1	348	36.5	gene_fragment	India	2008
DQ979320.1	615	48.3	gene_fragment	Canada	2023
DQ979321.2	846	36.9	gene_fragment	Canada	2023
DQ979322.1	861	41.7	gene_fragment	Canada	2023
DV018450.1	570	41.8	mRNA	Canada	2023
DV202937.1	496	51.6	mRNA	Canada	2023
EE572294.1	628	49.7	EST	Canada	2023
EE572295.1	1,155	32.2	EST	Canada	2023
EE572296.1	431	49.6	EST	Canada	2023
EE572297.1	484	49.6	EST	Canada	2023
EE572298.1	1,221	43.1	EST	Canada	2023
EE572299.1	705	48.9	EST	Canada	2023
EE572300.1	512	41.6	EST	Canada	2023
EE572301.1	734	48.8	EST	Canada	2023
EE572302.1	708	47.9	EST	Canada	2023
EE572303.1	665	49.3	EST	Canada	2023
EE572304.1	1,213	47.5	EST	Canada	2023
EE572305.1	585	51.3	EST	Canada	2023
EE572306.1	1,092	50.0	EST	Canada	2023
EE572307.1	1,190	46.0	EST	Canada	2023
EE572308.1	1,024	41.1	EST	Canada	2023
EE572309.1	1,146	47.0	EST	Canada	2023
EE572310.1	383	44.1	EST	Canada	2023
EE572311.1	464	48.1	EST	Canada	2023
EE572312.1	1,104	46.8	EST	Canada	2023
EE572313.1	1,031	45.2	EST	Canada	2023
EF078890.1	417	43.9	unknown	China	2001
EF093794.1	999	41.6	partial_sequence	Canada	2023
EF112453.1	1,380	40.0	unknown	Canada	2023
EF194079.1	615	48.5	gene_fragment	Thailand	1999

ACCESSION	LENGTH	GC	COMPLETENESS	COUNTRY	YEAR
EF197113.1	757	33.7	partial_sequence	Iran	2016
EF205278.1	486	44.0	partial_sequence	Vietnam	2006
EF205279.1	471	44.8	partial_sequence	Vietnam	2000
EF468498.1	356	47.8	unknown	India	2008
EF468499.1	986	32.7	unknown	India	2008
EF524222.1	870	44.4	unknown	Canada	2023
EF524223.1	1,200	38.8	unknown	Canada	2023
EF524224.1	885	39.1	unknown	Canada	2023
EF524225.1	966	42.0	unknown	Canada	2023
EF534251.1	846	36.8	gene_fragment	India	2007
EF534252.1	207	48.3	gene_fragment	India	2007
EF534253.1	615	43.1	gene_fragment	India	2007
EF534254.1	615	48.8	gene_fragment	India	2007
EF534255.1	1,401	40.4	gene_fragment	India	2007
EF564811.1	852	36.4	gene_fragment	India	2008
EF564812.1	1,401	40.4	gene_fragment	India	2008
EF661844.1	615	48.5	gene_fragment	India	2008
EF661845.1	843	37.2	partial_sequence	India	2008
ES490568.1	580	43.1	EST	Canada	2023
ES490569.1	709	43.6	EST	Canada	2023
ES490570.1	646	52.8	EST	Canada	2023
ES490571.1	598	47.7	EST	Canada	2023
ES490572.1	802	41.5	EST	Canada	2023
ES490573.1	914	43.3	EST	Canada	2023
ES490574.1	669	49.8	EST	Canada	2023
ES490575.1	1,012	50.7	EST	Canada	2023
ES490576.1	552	49.1	EST	Canada	2023
ES490577.1	695	52.2	EST	Canada	2023
ES490578.1	559	49.9	EST	Canada	2023
ES490579.1	640	45.6	EST	Canada	2023
ES490580.1	618	47.2	EST	Canada	2023
ES490581.1	479	50.9	EST	Canada	2023
ES490582.1	445	48.8	EST	Canada	2023
ES490583.1	550	46.0	EST	Canada	2023
ES490584.1	967	39.1	EST	Canada	2023
ES490585.1	521	40.1	EST	Canada	2023
ES490586.1	1,036	44.1	EST	Canada	2023
ES490587.1	311	44.0	EST	Canada	2023
ES490588.1	779	46.9	EST	Canada	2023
ES490589.1	461	49.2	EST	Canada	2023
ES490590.1	887	45.5	EST	Canada	2023
ES490591.1	663	51.0	EST	Canada	2023
ES490592.1	637	48.7	EST	Canada	2023
ES490593.1	1,430	32.4	EST	Canada	2023
ES490594.1	903	44.2	EST	Canada	2023
ES490595.1	910	43.0	EST	Canada	2023

ACCESSION	LENGTH	GC	COMPLETENESS	COUNTRY	YEAR
ES490596.1	959	44.8	EST	Canada	2023
ES490597.1	1,038	45.3	EST	Canada	2023
ES490598.1	562	50.7	EST	Canada	2023
ES490599.1	686	47.1	EST	Canada	2023
ES490600.1	454	45.4	EST	Canada	2023
ES490601.1	531	46.0	EST	Canada	2023
ES490602.1	546	44.9	EST	Canada	2023
ES490603.1	584	48.3	EST	Canada	2023
ES490604.1	970	51.4	EST	Canada	2023
ES490605.1	440	45.7	EST	Canada	2023
ES490606.1	457	47.7	EST	Canada	2023
ES490607.1	570	50.4	EST	Canada	2023
ES490608.1	864	51.6	EST	Canada	2023
ES490609.1	464	52.8	EST	Canada	2023
ES490610.1	653	47.2	EST	Canada	2023
ES490611.1	793	52.5	EST	Canada	2023
ES490612.1	980	50.3	EST	Canada	2023
ES490613.1	1,151	42.9	EST	Canada	2023
ES490614.1	906	42.2	EST	Canada	2023
ES490615.1	1,094	43.8	EST	Canada	2023
ES490616.1	1,142	38.5	EST	Canada	2023
ES490617.1	1,363	54.2	EST	Canada	2023
ES490618.1	1,090	44.3	EST	Canada	2023
ES490619.1	1,239	49.6	EST	Canada	2023
ES490620.1	1,169	46.5	EST	Canada	2023
ES490621.1	617	46.0	EST	Canada	2023
ES490622.1	423	44.9	EST	Canada	2023
ES490623.1	1,190	37.1	EST	Canada	2023
ES490624.1	858	44.8	EST	Canada	2023
ES490625.1	888	36.9	EST	Canada	2023
ES490626.1	631	39.0	EST	Canada	2023
ES490627.1	1,559	35.6	EST	Canada	2023
ES490628.1	1,461	34.6	EST	Canada	2023
ES490629.1	1,329	47.9	EST	Canada	2023
ES490630.1	1,208	47.0	EST	Canada	2023
ES490631.1	910	45.0	EST	Canada	2023
ES490632.1	748	43.2	EST	Canada	2023
ES490633.1	1,193	35.6	EST	Canada	2023
ES490634.1	409	46.0	EST	Canada	2023
ES490635.1	476	39.7	EST	Canada	2023
ES490636.1	792	49.9	EST	Canada	2023
ES490637.1	916	46.1	EST	Canada	2023
ES490638.1	900	44.1	EST	Canada	2023
ES490639.1	1,096	37.8	EST	Canada	2023
ES490640.1	900	37.2	EST	Canada	2023
ES490641.1	1,398	40.1	EST	Canada	2023

ACCESSION	LENGTH	GC	COMPLETENESS	COUNTRY	YEAR
ES490642.1	1,160	52.3	EST	Canada	2023
EU012447.1	366	51.1	gene_fragment	India	2007
EU327499.1	2,950	40.4	partial_sequence	India	2008
EU327500.1	2,364	39.3	gene_fragment	India	2008
EU327501.1	620	38.7	gene_fragment	India	2008
EU414753.1	615	48.6	gene_fragment	India	2008
EU414754.1	366	50.8	gene_fragment	India	2008
EU931451.2	702	47.1	partial_sequence	Mexico	2001
EU931452.1	960	39.1	gene_fragment	Mexico	2003
FJ001816.1	941	47.0	unknown	China	2001
FJ356711.1	680	46.0	partial_sequence	Thailand	2008
FJ356712.1	689	46.6	partial_sequence	Thailand	2008
FJ356713.1	689	45.3	partial_sequence	Thailand	2008
FJ356714.1	689	46.7	partial_sequence	Thailand	2008
FJ483831.1	3,864	42.0	gene_fragment	China	2007
FJ609650.1	570	41.8	partial_sequence	Mexico	2007
FJ756453.1	702	47.1	partial_sequence	Mexico	2005
FJ756454.1	702	47.0	partial_sequence	Mexico	2004
FJ756455.1	702	47.0	partial_sequence	Mexico	2004
FJ756456.1	702	47.1	partial_sequence	Mexico	2001
FJ756457.1	301	39.9	partial_sequence	Mexico	2005
FJ756458.1	301	39.9	partial_sequence	Mexico	2004
FJ756459.1	301	39.9	partial_sequence	Mexico	2004
FJ756460.1	301	39.9	partial_sequence	Mexico	2001
FJ756461.1	1,049	42.6	partial_sequence	Mexico	2005
FJ756462.1	1,049	42.6	partial_sequence	Mexico	2004
FJ756463.1	1,049	42.6	partial_sequence	Mexico	2004
FJ756464.1	1,049	42.6	partial_sequence	Mexico	2001
FJ756465.1	290	50.7	partial_sequence	Mexico	2005
FJ756466.1	290	50.7	partial_sequence	Mexico	2004
FJ756467.1	290	50.7	partial_sequence	Mexico	2004
FJ756468.1	290	50.7	partial_sequence	Mexico	2001
FJ756469.1	756	43.4	partial_sequence	Mexico	2005
FJ756470.1	756	43.4	partial_sequence	Mexico	2004
FJ756471.1	756	43.4	partial_sequence	Mexico	2004
FJ756472.1	756	43.4	partial_sequence	Mexico	2001
FJ756473.1	756	43.4	partial_sequence	Mexico	2001
FJ756474.1	921	39.7	partial_sequence	Mexico	2001
FJ756475.1	790	40.1	partial_sequence	Mexico	2001
FJ789570.1	930	38.3	partial_sequence	Mexico	2002
FN687469.1	540	51.7	partial_sequence	Canada	2023
FN687470.1	378	53.2	partial_sequence	Canada	2023
FN687471.1	216	53.2	partial_sequence	Canada	2023
GQ328028.1	366	50.8	gene_fragment	South Korea	2009
GQ328029.1	615	48.8	gene_fragment	South Korea	2009
GU265551.1	19	N/A	complete_genome	China	2009

ACCESSION	LENGTH	GC	COMPLETENESS	COUNTRY	YEAR
GU265552.1	21	N/A	complete_genome	China	2009
GU332708.1	1,044	47.2	unknown	Brazil	2005
GU734034.1	615	48.6	gene_fragment	India	2009
GU734035.1	366	51.1	gene_fragment	India	2009
HM484379.1	606	48.8	gene_fragment	India	2009
HM484380.1	606	48.8	gene_fragment	India	2009
HM484381.1	606	48.8	gene_fragment	India	2009
HM484382.1	606	48.8	gene_fragment	India	2009
HM484383.1	606	48.8	gene_fragment	India	2009
HM484384.1	606	48.8	gene_fragment	India	2009
HM484385.1	606	48.8	gene_fragment	India	2009
HM484386.1	606	48.8	gene_fragment	India	2009
HM484387.1	615	49.1	gene_fragment	India	2009
HM484388.1	615	49.1	gene_fragment	India	2009
HM484389.1	615	49.1	gene_fragment	India	2009
HM484390.1	615	49.1	gene_fragment	India	2009
HM778020.1	187	44.4	partial_sequence	India	2009
HQ007029.1	1,031	49.6	partial_sequence	Mexico	2001
HQ007030.1	761	48.4	partial_sequence	Mexico	2005
HQ007031.1	1,301	50.8	partial_sequence	Mexico	2004
HQ007032.1	1,355	50.9	partial_sequence	Mexico	2004
HQ007033.1	869	49.1	partial_sequence	Mexico	2001
HQ130032.1	525	49.9	partial_sequence	Brazil	2007
HQ257380.1	2,223	35.9	complete_genome	Mexico	2001
HQ257381.1	2,223	35.9	complete_genome	Mexico	2004
HQ257382.1	1,070	32.6	complete_genome	Mexico	2005
HQ257383.1	1,070	32.5	complete_genome	Mexico	2004
HQ257384.1	1,070	32.5	complete_genome	Mexico	2001
HQ286370.1	1,640	36.0	partial_sequence	Mexico	2001
HQ286371.1	1,640	36.0	partial_sequence	Mexico	2005
HQ735419.1	721	39.1	partial_sequence	Vietnam	2010
HV220405.1	555	43.6	gene_fragment	Canada	2023
HV220406.1	615	48.6	gene_fragment	Canada	2023
HV220407.1	25	N/A	gene_fragment	N/A	N/A
HV220408.1	25	N/A	gene_fragment	N/A	N/A
HV220409.1	615	43.1	gene_fragment	Canada	2023
HV220410.1	627	37.8	gene_fragment	Canada	2023
HV220411.1	25	N/A	gene_fragment	N/A	N/A
HV220412.1	29	N/A	gene_fragment	N/A	N/A
JN113093.1	22	N/A	complete_genome	China	2010
JN113094.1	22	N/A	complete_genome	China	2010
JN113095.1	22	N/A	complete_genome	China	2010
JN113096.1	22	N/A	complete_genome	China	2010
JN113097.1	22	N/A	complete_genome	China	2010
JN113098.1	22	N/A	complete_genome	China	2010
JN113099.1	22	N/A	complete_genome	China	2010

ACCESSION	LENGTH	GC	COMPLETENESS	COUNTRY	YEAR
JN113100.1	22	N/A	complete_genome	China	2010
JN113101.1	22	N/A	complete_genome	China	2010
JN113102.1	22	N/A	complete_genome	China	2010
JN113103.1	22	N/A	complete_genome	China	2010
JN113104.1	22	N/A	complete_genome	China	2010
JN113105.1	22	N/A	complete_genome	China	2010
JN113106.1	22	N/A	complete_genome	China	2010
JN113107.1	22	N/A	complete_genome	China	2010
JN113108.1	22	N/A	complete_genome	China	2010
JN113109.1	22	N/A	complete_genome	China	2010
JN113110.1	22	N/A	complete_genome	China	2010
JN113111.1	22	N/A	complete_genome	China	2010
JN113112.1	22	N/A	complete_genome	China	2010
JN113113.1	22	N/A	complete_genome	China	2010
JN113114.1	22	N/A	complete_genome	China	2010
JN113115.1	22	N/A	complete_genome	China	2010
JN113116.1	22	N/A	complete_genome	China	2010
JN113117.1	22	N/A	complete_genome	China	2010
JN113118.1	22	N/A	complete_genome	China	2010
JN113119.1	22	N/A	complete_genome	China	2010
JN113120.1	22	N/A	complete_genome	China	2010
JN113121.1	22	N/A	complete_genome	China	2010
JN113122.1	22	N/A	complete_genome	China	2010
JN113123.1	22	N/A	complete_genome	China	2010
JN113124.1	22	N/A	complete_genome	China	2010
JN113125.1	22	N/A	complete_genome	China	2010
JN113126.1	22	N/A	complete_genome	China	2010
JN113127.1	22	N/A	complete_genome	China	2010
JN113128.1	22	N/A	complete_genome	China	2010
JN113129.1	22	N/A	complete_genome	China	2010
JN113130.1	19	N/A	complete_genome	China	2010
JN113131.1	21	N/A	complete_genome	China	2010
JN113132.1	21	N/A	complete_genome	China	2010
JN165706.1	509	45.8	partial_sequence	India	2016
JN165707.1	534	47.0	partial_sequence	India	2016
JX027478.1	249	45.4	complete_genome	India	2008
JX027479.1	627	41.5	gene_fragment	India	2008
JX027480.1	366	51.1	gene_fragment	India	2008
JX027481.1	615	48.8	gene_fragment	India	2008
JX027482.1	786	36.9	unknown	India	2008
JX198549.1	508	45.7	partial_sequence	India	2012
JX444992.1	615	48.8	gene_fragment	Vietnam	2011
JX444993.1	615	48.8	gene_fragment	Vietnam	2012
JX444994.1	615	49.3	gene_fragment	Vietnam	2012
JX515788.1	295,884	41.0	complete_genome	South Korea	2011
JX564899.1	1,385	44.5	partial_sequence	Vietnam	2011

ACCESSION	LENGTH	GC	COMPLETENESS	COUNTRY	YEAR
KC189609.1	570	41.8	partial_sequence	Mexico	2008
KC189610.1	570	41.8	partial_sequence	Mexico	2008
KC189611.1	570	41.8	partial_sequence	Mexico	2008
KC417309.1	318	47.5	partial_sequence	Mexico	2006
KC484341.1	2,096	38.7	gene_fragment	China	2010
KC484342.1	619	38.6	gene_fragment	China	2010
KC906266.1	314	47.5	partial_sequence	Iran	2012
KC906267.1	729	54.6	partial_sequence	Iran	2010
KC906268.1	1,017	54.9	partial_sequence	Iran	2011
KC906269.1	510	48.2	partial_sequence	Iran	2010
KC906270.1	513	48.3	partial_sequence	Iran	2012
KC906271.1	504	48.2	partial_sequence	Iran	2012
KC906272.1	516	48.6	partial_sequence	Iran	2012
KC906273.1	411	49.1	partial_sequence	Iran	2012
KC906274.1	399	51.6	partial_sequence	Iran	2010
KC906275.1	249	51.0	partial_sequence	Iran	2011
KC906276.1	231	50.2	partial_sequence	Iran	2012
KC906277.1	216	50.0	partial_sequence	Iran	2012
KF032716.1	1,316	45.0	partial_sequence	Iran	2008
KF157831.1	339	54.3	partial_sequence	Iran	2012
KF157832.1	891	54.4	partial_sequence	Iran	2012
KF157833.1	939	54.5	partial_sequence	Iran	2012
KF157834.1	909	54.8	partial_sequence	Iran	2012
KF157835.1	408	53.4	partial_sequence	Iran	2012
KF157836.1	681	48.0	partial_sequence	Iran	2012
KF157837.1	618	48.9	partial_sequence	Iran	2012
KF157838.1	516	48.1	partial_sequence	Iran	2012
KF157839.1	1,143	54.4	partial_sequence	Iran	2012
KF157840.1	203	55.2	partial_sequence	Iran	2012
KF157841.1	210	51.4	partial_sequence	Iran	2012
KF157842.1	204	50.0	partial_sequence	Iran	2012
KF208686.1	1,343	45.5	unverified	Iran	2012
KF225732.1	583	47.2	partial_sequence	United States	2007
KF225733.1	5,043	38.5	unknown	United States	2007
KF225734.1	1,405	38.5	gene_fragment	United States	2007
KF225735.1	1,064	54.8	complete_genome	United States	2007
KF225736.1	697	45.9	gene_fragment	United States	2007
KF723558.1	612	48.5	partial_sequence	Iran	2011
KF771899.1	922	35.5	unknown	United States	1997
KF771900.1	751	46.6	unknown	United States	1997
KF771901.1	633	48.0	unknown	United States	1997
KF771902.1	2,225	37.1	unknown	United States	2002
KF771903.1	7,636	37.7	unknown	United States	2002
KF771904.1	6,439	38.0	unknown	United States	1997
KF771905.1	8,973	37.9	unknown	United States	1997
KF771906.1	763	36.3	unknown	United States	1997

ACCESSION	LENGTH	GC	COMPLETENESS	COUNTRY	YEAR
KF771907.1	8,531	37.8	unknown	United States	1997
KF771908.1	464	54.7	unknown	United States	1996
KF771909.1	5,124	38.1	unknown	United States	1996
KF835977.1	411	55.0	partial_sequence	Iran	2013
KF835978.1	276	51.1	partial_sequence	Iran	2013
KF835979.1	387	54.0	partial_sequence	Iran	2012
KF835980.1	684	52.3	partial_sequence	Iran	2012
KF844252.1	2,096	38.7	unverified	China	2010
KF922389.1	620	38.5	gene_fragment	China	2001
KF922390.1	620	38.5	gene_fragment	China	2001
KF922391.1	513	44.4	gene_fragment	China	2014
KF922392.1	519	44.3	gene_fragment	China	2014
KF922393.1	2,096	38.7	gene_fragment	China	2010
KF922394.1	2,096	38.8	gene_fragment	China	2010
KF922395.1	2,096	38.7	gene_fragment	China	2012
KF922396.1	451	53.2	gene_fragment	China	2010
KF922397.1	676	54.4	gene_fragment	China	2010
KF922398.1	541	53.2	gene_fragment	China	2011
KF922399.1	484	54.3	gene_fragment	China	2012
KF922400.1	721	48.1	gene_fragment	China	2010
KF922401.1	514	46.3	gene_fragment	China	2011
KF922402.1	859	49.0	gene_fragment	China	2011
KF922403.1	721	48.4	gene_fragment	China	2011
KF922404.1	790	48.5	gene_fragment	China	2012
KF922405.1	580	47.8	gene_fragment	China	2012
KF956790.1	660	53.6	partial_sequence	Iran	2013
KF956791.1	774	54.6	partial_sequence	Iran	2013
KF976715.1	560	46.2	partial_sequence	India	2011
KF976716.1	560	46.4	partial_sequence	Saudi Arabia	2009
KF981443.1	864	46.4	partial_sequence	Germany	2013
KJ719075.1	1,233	45.0	partial_sequence	Bangladesh	2013
KJ773998.1	1,365	45.4	partial_sequence	Bangladesh	2014
KJ773999.1	1,365	45.4	partial_sequence	Bangladesh	2014
KJ774000.1	1,365	45.4	partial_sequence	Bangladesh	2014
KJ774001.1	1,365	45.4	partial_sequence	Bangladesh	2014
KJ774002.1	1,365	45.4	partial_sequence	Bangladesh	2014
KJ774003.1	1,365	45.4	partial_sequence	Bangladesh	2014
KJ774004.1	1,365	45.4	partial_sequence	Bangladesh	2014
KJ774005.1	1,365	45.4	partial_sequence	Bangladesh	2014
KJ774006.1	1,365	45.4	partial_sequence	Bangladesh	2014
KJ774007.1	1,365	45.4	partial_sequence	Bangladesh	2014
KJ774008.1	1,365	45.4	partial_sequence	Bangladesh	2014
KJ774009.1	1,365	45.4	partial_sequence	Bangladesh	2014
KJ774010.1	1,365	45.4	partial_sequence	Bangladesh	2014
KJ774011.1	1,365	45.4	partial_sequence	Bangladesh	2014
KJ774012.1	1,365	45.4	partial_sequence	Bangladesh	2014

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KJ817413.1	516	50.2	partial_sequence	Bangladesh	2014
KJ817414.1	516	50.2	partial_sequence	Bangladesh	2014
KJ817415.1	516	50.2	partial_sequence	Bangladesh	2014
KJ817416.1	516	50.2	partial_sequence	Bangladesh	2014
KJ817417.1	516	50.2	partial_sequence	Bangladesh	2014
KJ817418.1	516	50.2	partial_sequence	Bangladesh	2014
KJ817419.1	516	50.2	partial_sequence	Bangladesh	2014
KJ817420.1	516	50.2	partial_sequence	Bangladesh	2014
KJ817421.1	969	39.8	partial_sequence	Bangladesh	2014
KJ817422.1	969	39.8	partial_sequence	Bangladesh	2014
KJ937964.1	359	40.7	partial_sequence	India	2013
KJ937965.1	474	44.9	partial_sequence	India	2013
KJ937966.1	581	38.5	partial_sequence	India	2013
KJ937967.1	361	40.4	partial_sequence	India	2013
KP219387.1	615	48.9	gene_fragment	Bangladesh	2014
KP219388.1	615	48.9	gene_fragment	Bangladesh	2014
KP455493.1	745	54.9	partial_sequence	Iran	2014
KP640278.1	683	52.6	partial_sequence	Iran	2014
KR057960.1	615	48.8	gene_fragment	India	2010
KR057961.1	615	48.8	gene_fragment	India	2014
KR083838.1	615	48.8	gene_fragment	Egypt	2014
KR083839.1	5,124	38.1	unknown	Egypt	2014
KR083840.1	464	55.0	unknown	Egypt	2014
KR083841.1	8,531	37.9	unknown	Egypt	2014
KR083842.1	763	36.4	unknown	Egypt	2014
KR083843.1	8,973	37.9	unknown	Egypt	2014
KR083844.1	6,439	37.9	unknown	Egypt	2014
KR083845.1	7,636	37.7	unknown	Egypt	2014
KR083846.1	2,225	37.1	unknown	Egypt	2014
KR083847.1	633	48.2	unknown	Egypt	2014
KR083848.1	751	46.6	unknown	Egypt	2014
KR083849.1	516	50.4	partial_sequence	Mozambique	2015
KR083850.1	516	50.4	partial_sequence	Mozambique	2015
KR083851.1	516	50.2	partial_sequence	Mozambique	2015
KR083852.1	516	50.4	partial_sequence	Mozambique	2015
KR083853.1	361	40.4	partial_sequence	Mozambique	2015
KR083854.1	633	48.0	unknown	Egypt	2014
KR083855.1	560	46.4	partial_sequence	Egypt	2014
KR083856.1	652	47.7	partial_sequence	N/A	2014
KR083857.1	774	54.6	partial_sequence	Egypt	2014
KR083858.1	1,064	54.8	gene_fragment	Egypt	2014
KR083859.1	2,096	38.7	gene_fragment	Egypt	2014
KR083860.1	2,096	38.7	gene_fragment	Egypt	2014
KR083861.1	366	51.1	gene_fragment	Egypt	2014
KR083862.1	660	53.6	partial_sequence	Egypt	2014
KR083863.1	969	39.8	partial_sequence	Egypt	2014

ACCESSION	LENGTH	GC	COMPLETENESS	COUNTRY	YEAR
KR083864.1	1,365	45.4	partial_sequence	Egypt	2014
KR083865.1	922	35.5	unknown	Egypt	2014
KT734624.1	761	48.2	partial_sequence	Mexico	2012
KT734625.1	1,010	54.8	partial_sequence	Mexico	2001
KT734626.1	1,010	54.8	partial_sequence	Mexico	2005
KT734627.1	1,010	54.8	partial_sequence	Mexico	2004
KT734628.1	1,010	54.8	partial_sequence	Mexico	2004
KT734629.1	1,037	54.6	partial_sequence	Mexico	2012
KT734630.1	852	48.9	partial_sequence	Mexico	2001
KT734631.1	852	49.3	partial_sequence	Mexico	2005
KT734632.1	782	49.1	partial_sequence	Mexico	2004
KT734633.1	782	49.1	partial_sequence	Mexico	2004
KT734634.1	645	47.6	partial_sequence	Mexico	2012
KT748521.1	871	43.3	partial_sequence	Mexico	2012
KT894028.3	941	47.0	partial_sequence	Iran	2010
KT894030.3	941	47.0	partial_sequence	Iran	2010
KT894031.3	941	47.0	partial_sequence	Iran	2011
KT894032.3	941	47.0	partial_sequence	Iran	2010
KT894040.3	941	47.0	partial_sequence	Iran	2012
KT923792.1	697	39.0	partial_sequence	Mexico	2001
KT923793.1	697	39.0	partial_sequence	Mexico	2005
KT923794.1	697	39.0	partial_sequence	Mexico	2004
KT923795.1	697	39.0	partial_sequence	Mexico	2004
KT923796.1	697	39.0	partial_sequence	Mexico	2012
KT923797.1	484	40.1	partial_sequence	Mexico	2001
KT923798.1	484	40.1	partial_sequence	Mexico	2005
KT923799.1	484	40.1	partial_sequence	Mexico	2004
KT923800.1	484	40.1	partial_sequence	Mexico	2004
KT923801.1	484	40.1	partial_sequence	Mexico	2012
KT923802.1	585	41.5	partial_sequence	Mexico	2001
KT923803.1	585	41.7	partial_sequence	Mexico	2005
KT923804.1	585	41.5	partial_sequence	Mexico	2004
KT923805.1	585	41.5	partial_sequence	Mexico	2004
KT923806.1	585	41.4	partial_sequence	Mexico	2012
KT923807.1	622	40.8	partial_sequence	Mexico	2001
KT923808.1	622	40.8	partial_sequence	Mexico	2005
KT923809.1	622	40.8	partial_sequence	Mexico	2004
KT923810.1	622	40.8	partial_sequence	Mexico	2004
KT923811.1	622	41.0	partial_sequence	Mexico	2012
KT957001.1	470	34.0	partial_sequence	Mexico	2001
KT957002.1	470	34.0	partial_sequence	Mexico	2005
KT957003.1	470	34.0	partial_sequence	Mexico	2004
KT957004.1	470	34.0	partial_sequence	Mexico	2004
KT957005.1	470	34.0	partial_sequence	Mexico	2012
KT957006.1	1,718	56.7	partial_sequence	Mexico	2001
KT957007.1	1,718	56.7	partial_sequence	Mexico	2005

ACCESSION	LENGTH	GC	COMPLETENESS	COUNTRY	YEAR
KT957008.1	1,718	56.7	partial_sequence	Mexico	2004
KT957009.1	1,718	56.7	partial_sequence	Mexico	2004
KT957010.1	1,934	56.4	partial_sequence	Mexico	2012
KT957011.1	831	34.5	gene_fragment	Mexico	2001
KT957012.1	831	34.5	gene_fragment	Mexico	2005
KT957013.1	831	34.5	gene_fragment	Mexico	2004
KT957014.1	831	34.5	gene_fragment	Mexico	2004
KT957015.1	831	34.5	gene_fragment	Mexico	2012
KT957016.1	459	40.3	gene_fragment	Mexico	2001
KT957017.1	459	40.3	gene_fragment	Mexico	2005
KT957018.1	459	40.3	gene_fragment	Mexico	2004
KT957019.1	459	40.3	gene_fragment	Mexico	2004
KT957020.1	459	40.3	gene_fragment	Mexico	2012
KT957021.1	610	38.5	partial_sequence	Mexico	2001
KT957022.1	610	38.5	partial_sequence	Mexico	2005
KT957023.1	610	38.5	partial_sequence	Mexico	2004
KT957024.1	610	38.5	partial_sequence	Mexico	2004
KT957025.1	610	38.5	partial_sequence	Mexico	2012
KT957026.1	1,606	39.0	partial_sequence	Mexico	2001
KT957027.1	1,606	39.0	partial_sequence	Mexico	2005
KT957028.1	1,606	39.0	partial_sequence	Mexico	2004
KT957029.1	1,606	39.0	partial_sequence	Mexico	2004
KT957030.1	1,603	38.9	partial_sequence	Mexico	2012
KT957031.1	763	40.8	partial_sequence	Mexico	2001
KT957032.1	763	40.8	partial_sequence	Mexico	2005
KT957033.1	763	40.8	partial_sequence	Mexico	2004
KT957034.1	763	40.8	partial_sequence	Mexico	2004
KT957035.1	760	40.7	partial_sequence	Mexico	2012
KT957036.1	871	37.7	partial_sequence	Mexico	2001
KT957037.1	871	37.7	partial_sequence	Mexico	2005
KT957038.1	871	37.7	partial_sequence	Mexico	2004
KT957039.1	871	37.7	partial_sequence	Mexico	2004
KT957040.1	871	37.7	partial_sequence	Mexico	2012
KT957041.1	779	35.2	partial_sequence	Mexico	2001
KT957042.1	779	35.2	partial_sequence	Mexico	2005
KT957043.1	779	35.2	partial_sequence	Mexico	2004
KT957044.1	779	35.2	partial_sequence	Mexico	2004
KT957045.1	779	35.2	partial_sequence	Mexico	2012
KT957046.1	777	38.2	partial_sequence	Mexico	2001
KT957047.1	777	38.2	partial_sequence	Mexico	2005
KT957048.1	777	38.4	partial_sequence	Mexico	2004
KT957049.1	777	38.4	partial_sequence	Mexico	2004
KT957050.1	777	38.2	partial_sequence	Mexico	2012
KT957051.1	521	39.9	partial_sequence	Mexico	2001
KT957052.1	521	39.9	partial_sequence	Mexico	2005
KT957053.1	521	39.9	partial_sequence	Mexico	2004

ACCESSION	LENGTH	GC	COMPLETENESS	COUNTRY	YEAR
KT957054.1	521	39.9	partial_sequence	Mexico	2004
KT957055.1	521	39.9	partial_sequence	Mexico	2012
KT957056.1	482	39.4	gene_fragment	Mexico	2001
KT957057.1	482	39.4	gene_fragment	Mexico	2005
KT957058.1	473	39.3	gene_fragment	Mexico	2004
KT957059.1	473	39.3	gene_fragment	Mexico	2004
KT957060.1	482	39.4	gene_fragment	Mexico	2012
KT957061.1	842	36.2	partial_sequence	Mexico	2001
KT957062.1	842	36.2	partial_sequence	Mexico	2005
KT957063.1	842	36.2	partial_sequence	Mexico	2004
KT957064.1	842	36.2	partial_sequence	Mexico	2004
KT957065.1	842	36.2	partial_sequence	Mexico	2012
KT957066.1	862	38.3	partial_sequence	Mexico	2001
KT957067.1	862	38.3	partial_sequence	Mexico	2005
KT957068.1	862	38.3	partial_sequence	Mexico	2004
KT957069.1	862	38.3	partial_sequence	Mexico	2004
KT957070.1	862	38.3	partial_sequence	Mexico	2012
KT957071.1	285	41.4	partial_sequence	Mexico	2001
KT957072.1	285	41.4	partial_sequence	Mexico	2005
KT957073.1	285	41.4	partial_sequence	Mexico	2004
KT957074.1	285	41.4	partial_sequence	Mexico	2004
KT957075.1	285	41.4	partial_sequence	Mexico	2012
KT957076.1	983	41.6	partial_sequence	Mexico	2001
KT957077.1	983	41.6	partial_sequence	Mexico	2005
KT957078.1	983	41.6	partial_sequence	Mexico	2004
KT957079.1	983	41.6	partial_sequence	Mexico	2004
KT957080.1	983	41.5	partial_sequence	Mexico	2012
KT963933.1	615	48.8	gene_fragment	India	2009
KT963934.1	557	48.6	partial_sequence	India	2008
KT963935.1	608	49.0	partial_sequence	India	2009
KT963936.1	539	48.6	partial_sequence	India	2009
KT963937.1	615	48.8	gene_fragment	India	2009
KT963938.1	615	48.8	gene_fragment	India	2009
KT963939.1	615	48.9	gene_fragment	India	2009
KT963940.1	615	48.8	gene_fragment	India	2009
KT963941.1	556	48.7	partial_sequence	India	2009
KT963942.1	615	48.8	partial_sequence	India	2009
KT963943.1	615	48.6	gene_fragment	India	2009
KT963944.1	615	48.8	gene_fragment	India	2009
KT963945.1	541	50.1	partial_sequence	India	2009
KT964168.1	369	46.6	partial_sequence	Mexico	2001
KT964169.1	369	46.6	partial_sequence	Mexico	2005
KT964170.1	369	46.6	partial_sequence	Mexico	2004
KT964171.1	369	46.6	partial_sequence	Mexico	2004
KT964172.1	369	46.6	partial_sequence	Mexico	2012
KT964173.1	1,233	46.1	partial_sequence	Mexico	2001

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KT964174.1	1,233	46.1	partial_sequence	Mexico	2005
KT964175.1	1,233	46.1	partial_sequence	Mexico	2004
KT964176.1	1,233	46.1	partial_sequence	Mexico	2004
KT964177.1	1,233	46.1	partial_sequence	Mexico	2012
KT964178.1	2,298	44.1	partial_sequence	Mexico	2001
KT964179.1	2,298	44.1	partial_sequence	Mexico	2005
KT964180.1	2,298	44.1	partial_sequence	Mexico	2004
KT964181.1	2,298	44.1	partial_sequence	Mexico	2004
KT964182.1	2,298	44.1	partial_sequence	Mexico	2012
KT964183.1	1,594	44.8	partial_sequence	Mexico	2001
KT964184.1	1,594	44.9	partial_sequence	Mexico	2005
KT964185.1	1,594	44.8	partial_sequence	Mexico	2004
KT964186.1	1,594	44.8	partial_sequence	Mexico	2004
KT964187.1	1,594	44.8	partial_sequence	Mexico	2012
KT964188.1	935	44.4	partial_sequence	Mexico	2001
KT964189.1	935	44.4	partial_sequence	Mexico	2005
KT964190.1	935	44.4	partial_sequence	Mexico	2004
KT964191.1	935	44.4	partial_sequence	Mexico	2004
KT964192.1	935	44.5	partial_sequence	Mexico	2012
KT964193.1	1,709	43.9	partial_sequence	Mexico	2001
KT964194.1	1,709	43.9	partial_sequence	Mexico	2005
KT964195.1	1,709	43.9	partial_sequence	Mexico	2004
KT964196.1	1,709	43.9	partial_sequence	Mexico	2004
KT964197.1	1,709	44.0	partial_sequence	Mexico	2012
KT964198.1	976	38.2	partial_sequence	Mexico	2001
KT964199.1	976	38.2	partial_sequence	Mexico	2005
KT964200.1	976	38.2	partial_sequence	Mexico	2004
KT964201.1	976	38.2	partial_sequence	Mexico	2004
KT964202.1	976	38.2	partial_sequence	Mexico	2012
KT964203.1	428	34.6	partial_sequence	Mexico	2001
KT964204.1	428	34.6	partial_sequence	Mexico	2005
KT964205.1	428	34.6	partial_sequence	Mexico	2004
KT964206.1	428	34.6	partial_sequence	Mexico	2004
KT964207.1	428	34.6	partial_sequence	Mexico	2012
KT964208.1	986	38.6	partial_sequence	Mexico	2005
KT964209.1	986	38.6	partial_sequence	Mexico	2004
KT964210.1	986	38.6	partial_sequence	Mexico	2004
KT964211.1	986	38.6	partial_sequence	Mexico	2012
KT964212.1	743	41.0	gene_fragment	Mexico	2012
KT964213.1	432	42.1	partial_sequence	Mexico	2012
KT964214.1	1,007	41.9	partial_sequence	Mexico	2012
KT964215.1	691	43.0	partial_sequence	Mexico	2012
KT964216.1	1,700	37.3	partial_sequence	Mexico	2012
KT964217.1	1,963	46.0	partial_sequence	Mexico	2005
KT964218.1	2,000	46.2	partial_sequence	Mexico	2004
KT964219.1	2,000	46.2	partial_sequence	Mexico	2004

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KT964220.1	1,963	46.0	partial_sequence	Mexico	2012
KT995470.1	294,261	41.0	complete_genome	China	2010
KT995471.1	284,148	41.0	complete_genome	China	2010
KT995472.1	309,286	40.9	complete_genome	China	1994
KU216744.2	300,087	41.0	complete_genome	Mexico	2008
KU359256.1	400	46.5	partial_sequence	China	2015
KU531723.1	319	48.6	partial_sequence	Saudi Arabia	2014
KU556686.1	615	48.6	EST	India	2015
KU556687.1	641	42.6	EST	India	2015
KU556688.1	627	38.1	gene_fragment	India	2015
KU556689.1	366	51.1	gene_fragment	India	2015
KU746815.1	963	39.2	gene_fragment	China	2014
KU746816.1	943	43.0	gene_fragment	China	2014
KU746817.1	2,720	38.5	gene_fragment	China	2014
KX388150.1	627	41.5	gene_fragment	India	2008
KX492915.2	941	47.0	partial_sequence	Iran	2015
KX501220.1	606	37.6	gene_fragment	Thailand	2012
KX501221.1	527	36.0	gene_fragment	Thailand	2012
KX501222.1	1,743	38.1	gene_fragment	Thailand	2012
KX501223.1	1,634	38.1	gene_fragment	Thailand	2012
KX584738.3	941	47.0	partial_sequence	Iran	2015
KX584739.3	941	47.0	partial_sequence	Iran	2015
KX584741.1	717	52.6	partial_sequence	Iran	2013
KX650068.2	401	55.4	partial_sequence	Iran	2010
KX686117.1	300,223	41.0	complete_genome	China	2015
KX694234.2	750	55.1	partial_sequence	Iran	2013
KX694235.2	729	54.6	partial_sequence	Iran	2013
KX694236.2	734	54.5	partial_sequence	Iran	2014
KX694237.2	734	54.5	partial_sequence	Iran	2014
KX694238.2	734	54.5	partial_sequence	Iran	2014
KX694239.2	720	55.3	partial_sequence	Iran	2014
KX694240.2	784	54.5	partial_sequence	Iran	2013
KX694241.2	739	55.1	partial_sequence	Iran	2013
KX694242.2	739	54.8	partial_sequence	Iran	2013
KX714112.3	941	47.0	partial_sequence	Iran	2015
KX714113.3	941	47.0	partial_sequence	Iran	2015
KX980155.1	555	41.6	partial_sequence	India	2016
KX981593.1	607	47.0	partial_sequence	India	2014
KY273305.1	244	50.0	partial_sequence	Philippines	2014
KY273306.1	244	50.0	partial_sequence	Philippines	2014
KY273307.1	244	50.0	partial_sequence	Philippines	2014
KY273308.1	244	50.0	partial_sequence	Philippines	2014
KY273309.1	244	50.0	partial_sequence	Philippines	2014
KY273310.1	244	50.0	partial_sequence	Philippines	2014
KY273311.1	244	50.0	partial_sequence	Philippines	2014
KY273312.1	244	50.0	partial_sequence	Philippines	2014

ACCESSION	LENGTH	GC	COMPLETENESS	COUNTRY	YEAR
KY273313.1	244	50.0	partial_sequence	Philippines	2014
KY273314.1	244	50.0	partial_sequence	Philippines	2014
KY273315.1	244	50.0	partial_sequence	Philippines	2014
KY273316.1	244	50.0	partial_sequence	Philippines	2014
KY273317.1	244	50.0	partial_sequence	Philippines	2014
KY273318.1	244	50.0	partial_sequence	Philippines	2014
KY273319.1	244	50.0	partial_sequence	Philippines	2014
KY273320.1	244	50.0	partial_sequence	Philippines	2014
KY273321.1	244	50.0	partial_sequence	Philippines	2014
KY273322.1	244	50.0	partial_sequence	Philippines	2014
KY273323.1	244	50.0	partial_sequence	Philippines	2014
KY273324.1	244	50.0	partial_sequence	Philippines	2014
KY273325.1	244	50.0	partial_sequence	Philippines	2014
KY273326.1	244	50.0	partial_sequence	Philippines	2014
KY273327.1	244	50.0	partial_sequence	Philippines	2014
KY273328.1	244	50.0	partial_sequence	Philippines	2014
KY273329.1	244	50.0	partial_sequence	Philippines	2015
KY273330.1	244	50.0	partial_sequence	Philippines	2015
KY273331.1	244	50.0	partial_sequence	Philippines	2015
KY273332.1	244	50.0	partial_sequence	Philippines	2015
KY273333.1	244	50.0	partial_sequence	Philippines	2015
KY273334.1	244	50.0	partial_sequence	Philippines	2015
KY273335.1	244	50.0	partial_sequence	Philippines	2015
KY273336.1	244	50.0	partial_sequence	Philippines	2015
KY273337.1	244	50.0	partial_sequence	Philippines	2015
KY273338.1	244	50.0	partial_sequence	Philippines	2015
KY273339.1	244	50.0	partial_sequence	Philippines	2015
KY273340.1	244	50.0	partial_sequence	Philippines	2015
KY273341.1	720	49.2	partial_sequence	Philippines	2014
KY273342.1	506	49.0	partial_sequence	Philippines	2014
KY273343.1	721	49.2	partial_sequence	Philippines	2014
KY273344.1	505	49.1	partial_sequence	Philippines	2014
KY273345.1	710	49.3	partial_sequence	Philippines	2014
KY273346.1	659	49.8	partial_sequence	Philippines	2014
KY273347.1	720	49.2	partial_sequence	Philippines	2014
KY273348.1	508	48.8	partial_sequence	Philippines	2014
KY273349.1	668	50.5	partial_sequence	Philippines	2014
KY273350.1	459	50.3	partial_sequence	Philippines	2014
KY273351.1	720	49.3	partial_sequence	Philippines	2014
KY273352.1	669	49.6	partial_sequence	Philippines	2014
KY273353.1	669	49.8	partial_sequence	Philippines	2014
KY273354.1	515	48.4	partial_sequence	Philippines	2014
KY273355.1	661	49.6	partial_sequence	Philippines	2014
KY273356.1	507	49.1	partial_sequence	Philippines	2014
KY273357.1	667	49.8	partial_sequence	Philippines	2014
KY273358.1	500	50.2	partial_sequence	Philippines	2015

ACCESSION	LENGTH	GC	COMPLETENESS	COUNTRY	YEAR
KY273359.1	453	48.6	partial_sequence	Philippines	2015
KY273360.1	429	49.0	partial_sequence	Philippines	2015
KY273361.1	566	52.1	partial_sequence	Philippines	2015
KY273362.1	404	52.2	partial_sequence	Philippines	2015
KY273363.1	404	52.2	partial_sequence	Philippines	2015
KY273364.1	566	52.1	partial_sequence	Philippines	2015
KY273365.1	566	52.1	partial_sequence	Philippines	2015
KY273366.1	404	52.2	partial_sequence	Philippines	2015
KY474512.1	915	46.6	partial_sequence	Iran	2016
KY786111.1	876	38.7	partial_sequence	Egypt	2016
KY827813.1	281,054	41.0	complete_genome	China	2012
LG144569.1	720	50.1	unknown	Canada	2023
LG144570.1	450	52.4	unknown	Canada	2023
LY566065.1	20	N/A	unknown	N/A	N/A
LY566066.1	20	N/A	unknown	N/A	N/A
MF101757.1	1,311	45.5	partial_sequence	China	2016
MF161440.1	888	45.5	partial_sequence	Australia	2017
MF161441.1	1,284	45.5	partial_sequence	Australia	2017
MF161442.1	1,284	45.5	partial_sequence	Australia	2017
MF161443.1	1,284	45.5	partial_sequence	Australia	2017
MF287678.1	786	45.8	partial_sequence	Australia	2017
MF348242.1	786	45.7	partial_sequence	Australia	2017
MF405922.1	650	48.3	partial_sequence	Australia	2017
MF489075.1	580	49.5	partial_sequence	Bangladesh	2015
MF489076.1	615	48.8	gene_fragment	Bangladesh	2015
MF489077.1	567	48.5	partial_sequence	Bangladesh	2015
MF489078.1	549	48.6	partial_sequence	Bangladesh	2017
MF489079.1	552	48.5	partial_sequence	Bangladesh	2017
MF489080.1	375	49.9	partial_sequence	Bangladesh	2017
MF489081.1	558	48.4	partial_sequence	Bangladesh	2017
MF489082.1	549	48.3	partial_sequence	Bangladesh	2017
MF537411.1	786	45.8	partial_sequence	Australia	2017
MF682099.1	237	48.1	unverified	Australia	2017
MF682100.1	920	45.4	unverified	Australia	2017
MF682101.1	642	46.0	unverified	Australia	2017
MF682102.1	573	53.2	unverified	Australia	2017
MF682103.1	94	N/A	unverified	Australia	2017
MF682104.1	474	50.2	unverified	Australia	2017
MF682105.1	225	48.0	unverified	Vietnam	2017
MF682106.1	935	45.4	unverified	Vietnam	2017
MF682107.1	654	45.6	unverified	Vietnam	2017
MF682108.1	567	52.9	unverified	Vietnam	2017
MF682109.1	336	48.2	unverified	Vietnam	2017
MF682110.1	786	45.8	partial_sequence	Australia	2017
MF682111.1	786	45.7	partial_sequence	Australia	2017
MF682112.1	786	45.7	unverified	Vietnam	2017

ACCESSION	LENGTH	GC	COMPLETENESS	COUNTRY	YEAR
MF682113.1	213	48.4	unverified	Vietnam	2017
MF682114.1	645	46.0	unverified	Vietnam	2017
MF682115.1	528	51.9	unverified	Vietnam	2017
MF682116.1	85	N/A	unverified	Vietnam	2017
MF682117.1	336	48.5	unverified	Vietnam	2017
MF682118.1	225	48.4	unverified	China	2017
MF682119.1	920	45.4	unverified	China	2017
MF682120.1	588	47.1	unverified	China	2017
MF682121.1	556	53.1	unverified	China	2017
MF682122.1	474	50.0	unverified	China	2017
MF682123.1	195	47.7	unverified	Vietnam	2017
MF682124.1	916	45.6	unverified	Vietnam	2017
MF682125.1	528	47.0	unverified	Vietnam	2017
MF682126.1	399	50.4	unverified	Vietnam	2017
MF682127.1	336	48.5	unverified	Vietnam	2017
MF768985.1	285,973	41.0	complete_genome	Australia	2016
MF784752.1	292,912	41.0	partial_sequence	Brazil	2015
MG099767.1	1,922	46.1	unknown	Brazil	2015
MG099768.1	3,592	43.0	unknown	Brazil	2015
MG099769.1	1,706	44.9	unknown	Brazil	2015
MG099770.1	761	48.1	unknown	Brazil	2015
MG099771.1	983	46.4	unknown	Brazil	2015
MG099772.1	765	46.9	unknown	Brazil	2015
MG099773.1	1,756	44.8	unknown	Brazil	2015
MG099774.1	1,519	45.6	unknown	Brazil	2015
MG099775.1	2,116	44.5	unknown	Brazil	2015
MG264599.1	314,232	41.0	complete_genome	Brazil	2015
MG432474.1	263,174	41.1	partial_sequence	Mexico	2011
MG432475.1	289,870	41.0	partial_sequence	Mexico	2012
MG432476.1	277,297	41.0	partial_sequence	Mexico	2012
MG432477.1	290,879	41.0	partial_sequence	Mexico	2011
MG432478.1	272,607	40.8	partial_sequence	Mexico	2005
MG432479.1	283,858	41.1	partial_sequence	Mexico	2011
MG432480.1	270,274	41.1	partial_sequence	Mexico	2011
MG432481.1	278,223	41.1	partial_sequence	Mexico	2011
MG432482.1	257,675	41.2	partial_sequence	Mexico	2012
MG702567.1	280,591	40.9	complete_genome	India	2013
MH090824.1	288,997	41.0	complete_genome	Ecuador	2015
MH268228.1	755	45.6	partial_sequence	India	2018
MH365436.1	255	44.7	partial_sequence	India	2017
MH425505.1	615	48.8	gene_fragment	Saudi Arabia	2018
MH425506.1	615	48.8	gene_fragment	Saudi Arabia	2018
MH425507.1	614	43.2	complete_genome	Saudi Arabia	2018
MH425508.1	613	43.4	complete_genome	Saudi Arabia	2018
MH553567.1	308	36.4	partial_sequence	Costa Rica	2017
MH757094.1	342	43.0	partial_sequence	India	2017

ACCESSION	LENGTH	GC	COMPLETENESS	COUNTRY	YEAR
MH757095.1	255	45.1	partial_sequence	India	2017
MH757096.1	276	44.9	partial_sequence	India	2017
MH883318.1	270,609	41.7	partial_sequence	India	2017
MH883319.1	276,835	41.0	partial_sequence	India	2012
MK239903.1	1,261	36.3	unverified	Brazil	2015
MK239904.1	1,253	36.5	unverified	Brazil	2015
MK239905.1	661	35.0	unverified	Brazil	2015
MK239906.1	1,844	40.2	unverified	Brazil	2015
MK239907.1	600	55.3	unverified	Brazil	2015
MK239908.1	362	50.5	unverified	Brazil	2015
MK239909.1	471	51.6	unverified	Brazil	2015
MK239910.1	793	52.5	unverified	Brazil	2015
MK239911.1	902	52.4	unverified	Brazil	2015
MK239912.1	686	52.2	unverified	Brazil	2015
MK239913.1	826	45.9	unverified	Brazil	2015
MK239914.1	472	50.6	unverified	Brazil	2015
MK560209.1	409	40.3	unverified	N/A	2019
MK560210.1	398	40.0	unverified	N/A	2019
MK752931.1	941	47.0	partial_sequence	India	2019
MN078153.1	301	40.2	partial_sequence	Thailand	2020
MN078154.1	301	40.2	partial_sequence	Thailand	2020
MN462691.1	773	52.0	partial_sequence	Sri Lanka	2014
MN462692.1	773	52.0	partial_sequence	Sri Lanka	2014
MN462693.1	773	51.8	partial_sequence	Sri Lanka	2015
MN462694.1	773	52.0	partial_sequence	Sri Lanka	2016
MN462695.1	773	52.0	partial_sequence	Sri Lanka	2016
MN462696.1	449	51.5	partial_sequence	Sri Lanka	2016
MN462697.1	611	51.7	partial_sequence	Sri Lanka	2016
MN462698.1	611	51.7	partial_sequence	Sri Lanka	2016
MN462699.1	611	51.7	partial_sequence	Sri Lanka	2016
MN462700.1	611	51.7	partial_sequence	Sri Lanka	2016
MN475881.1	771	40.1	unverified	Canada	2023
MN481520.1	936	46.7	partial_sequence	Vietnam	2019
MN625272.1	675	42.8	gene_fragment	Iran	2018
MN840357.1	305,094	41.0	complete_genome	United States	2017
MT108485.1	878	46.6	partial_sequence	Iran	2018
MT108486.1	878	46.6	partial_sequence	Iran	2018
MT108487.1	878	46.6	partial_sequence	Iran	2018
MT363786.1	456	50.7	partial_sequence	Bangladesh	2018
MT363787.1	432	50.2	partial_sequence	Bangladesh	2018
MT363788.1	456	50.7	partial_sequence	Bangladesh	2018
MT363789.1	456	50.7	partial_sequence	Bangladesh	2018
MT363790.1	456	50.7	partial_sequence	Bangladesh	2018
MT363791.1	456	50.7	partial_sequence	Bangladesh	2018
MT363792.1	456	50.7	partial_sequence	Bangladesh	2018
MT363793.1	456	50.7	partial_sequence	Bangladesh	2018

ACCESSION	LENGTH	GC	COMPLETENESS	COUNTRY	YEAR
MT363794.1	456	50.7	partial_sequence	Bangladesh	2018
MT363795.1	414	50.2	partial_sequence	Bangladesh	2018
MT363796.1	456	50.7	partial_sequence	Bangladesh	2018
MT363797.1	309	50.8	partial_sequence	Bangladesh	2018
MT363798.1	456	50.7	partial_sequence	Bangladesh	2018
MT363799.1	456	50.7	partial_sequence	Bangladesh	2018
MT363800.1	456	50.7	partial_sequence	Bangladesh	2018
MT543321.1	830	46.9	partial_sequence	South Korea	2020
MT543322.1	851	46.4	partial_sequence	South Korea	2020
MT666093.1	756	45.4	partial_sequence	Brazil	2014
MT755013.1	745	46.2	partial_sequence	Norway	2015
MT896428.1	837	43.5	unverified	China	2013
MW248106.1	285,601	41.1	partial_sequence	India	2019
MW248107.1	281,813	41.1	partial_sequence	India	2018
MW248108.1	281,160	41.0	partial_sequence	India	2016
MW498975.1	851	46.4	partial_sequence	South Korea	2020
MW498976.1	855	46.5	partial_sequence	South Korea	2018
MW625872.1	183	49.2	partial_sequence	China	2016
MW625873.1	183	49.2	partial_sequence	China	2016
MW625874.1	183	49.2	partial_sequence	China	2016
MW625875.1	183	49.2	partial_sequence	China	2016
MW625876.1	183	49.2	partial_sequence	China	2016
MZ098151.1	570	41.8	partial_sequence	India	2020
MZ327619.1	721	48.4	partial_sequence	Madagascar	2012
MZ327620.1	652	47.9	partial_sequence	Madagascar	2012
MZ327621.1	583	47.0	partial_sequence	Madagascar	2014
MZ327622.1	792	48.9	partial_sequence	Madagascar	2016
MZ327623.1	348	37.6	partial_sequence	Madagascar	2012
MZ327624.1	1,739	41.8	partial_sequence	Madagascar	2012
MZ327625.1	1,850	38.4	partial_sequence	Madagascar	2012
MZ327626.1	1,265	36.8	partial_sequence	Madagascar	2012
MZ383193.1	525	49.9	partial_sequence	Bangladesh	2019
MZ383194.1	554	48.7	partial_sequence	Bangladesh	2019
MZ383195.1	498	50.2	partial_sequence	Bangladesh	2018
MZ383196.1	615	48.9	gene_fragment	Bangladesh	2018
MZ383197.1	615	48.9	gene_fragment	Bangladesh	2018
MZ383198.1	615	48.9	gene_fragment	Bangladesh	2018
MZ383199.1	554	48.7	partial_sequence	Bangladesh	2018
MZ501375.1	859	46.0	partial_sequence	Iran	2020
NC_003225.3	309,286	40.9	complete_genome	China	1994
NC_075105.1	305,119	41.0	complete_genome	China	2001
OM350393.2	278,607	40.8	partial_sequence	Peru	2020
ON457602.1	292,172	41.0	partial_sequence	South Korea	2018
OQ657458.1	293,023	41.0	complete_genome	Madagascar	2015
OQ858617.1	285,974	41.2	complete_genome	United States	2023
OR636681.1	273,795	40.7	complete_genome	South Africa	2022

ACCESSION	LENGTH	GC	COMPLETENESS	COUNTRY	YEAR
PP057867.1	41,905	41.7	unverified	Bangladesh	2021
PP057868.1	40,695	40.7	unverified	Bangladesh	2021
PP057869.1	19,623	42.5	unverified	Bangladesh	2021
PP057870.1	34,261	42.3	unverified	Bangladesh	2021
PP057871.1	23,383	40.1	unverified	Bangladesh	2021
PP057872.1	21,089	40.7	unverified	Bangladesh	2021
PP057873.1	18,227	40.4	unverified	Bangladesh	2021
PP057874.1	15,891	41.7	unverified	Bangladesh	2021
PP057875.1	15,012	40.0	unverified	Bangladesh	2021
PP057876.1	9,687	39.1	unverified	Bangladesh	2021
PP057877.1	8,730	41.2	unverified	Bangladesh	2021
PP057878.1	6,540	41.0	unverified	Bangladesh	2021
PP057879.1	6,959	41.0	unverified	Bangladesh	2021
PP057880.1	6,188	41.2	unverified	Bangladesh	2021
PP057881.1	3,962	41.4	unverified	Bangladesh	2021
PP057882.1	5,193	39.8	unverified	Bangladesh	2021
PP057883.1	424	47.4	unverified	Bangladesh	2021
PP057884.1	432	62.5	unverified	Bangladesh	2021
PP057885.1	353	45.0	unverified	Bangladesh	2021
PP057886.1	291	45.4	unverified	Bangladesh	2021
PP057887.1	286	45.1	unverified	Bangladesh	2021
PP057888.1	180	46.7	unverified	Bangladesh	2021
PP057889.1	277,275	41.1	unverified	Bangladesh	2021
PP057890.1	15,980	29.2	unverified	Bangladesh	2021
PP057891.1	12,482	44.4	unverified	Bangladesh	2021
PP057892.1	7,661	38.9	unverified	Bangladesh	2021
PP057893.1	7,435	53.8	unverified	Bangladesh	2021
PP057894.1	6,185	46.8	unverified	Bangladesh	2021
PP057895.1	6,161	62.4	unverified	Bangladesh	2021
PP057896.1	5,943	57.3	unverified	Bangladesh	2021
PP057897.1	5,434	43.8	unverified	Bangladesh	2021
PP057898.1	5,377	40.5	unverified	Bangladesh	2021
PP057899.1	5,360	40.7	unverified	Bangladesh	2021
PP057900.1	4,452	41.5	unverified	Bangladesh	2021
PP057901.1	4,181	62.4	unverified	Bangladesh	2021
PP057902.1	2,853	46.3	unverified	Bangladesh	2021
PP057903.1	2,778	44.7	unverified	Bangladesh	2021
PP057904.1	2,433	47.1	unverified	Bangladesh	2021
PP057905.1	1,947	42.8	unverified	Bangladesh	2021
PP057906.1	1,302	51.6	unverified	Bangladesh	2021
PP057907.1	1,256	51.2	unverified	Bangladesh	2021
PP057908.1	1,240	32.7	unverified	Bangladesh	2021
PP057909.1	1,052	15.0	unverified	Bangladesh	2021
PP057910.1	241	49.8	unverified	Bangladesh	2021
PP057911.1	156,552	41.4	unverified	Bangladesh	2023
PP057912.1	74,872	40.8	unverified	Bangladesh	2023

ACCESSION	LENGTH	GC	COMPLETENESS	COUNTRY	YEAR
PP057913.1	25,327	41.1	unverified	Bangladesh	2023
PP057914.1	20,573	39.9	unverified	Bangladesh	2023
PP057915.1	15,983	29.3	unverified	Bangladesh	2023
PP057916.1	11,389	60.8	unverified	Bangladesh	2023
PP057917.1	3,580	56.4	unverified	Bangladesh	2023
PP057918.1	1,008	47.3	unverified	Bangladesh	2023
PP057919.1	286	40.6	unverified	Bangladesh	2023
PP057920.1	257	46.3	unverified	Bangladesh	2023
PP057921.1	255	46.3	unverified	Bangladesh	2023
PP057922.1	254	46.5	unverified	Bangladesh	2023
PP057923.1	253	45.9	unverified	Bangladesh	2023
PP057924.1	251	45.0	unverified	Bangladesh	2023
PP057925.1	221,305	41.3	unverified	Bangladesh	2023
PP057926.1	57,160	40.5	unverified	Bangladesh	2023
PP057927.1	15,983	29.3	unverified	Bangladesh	2023
PP057928.1	251	45.0	unverified	Bangladesh	2023
PP128489.1	288,217	41.1	complete_genome	Japan	2019
PP128490.1	279,886	41.1	complete_genome	Japan	2020
PP128491.1	288,229	41.1	complete_genome	Japan	2020
PP134839.1	309,286	40.9	complete_genome	Bangladesh	2023
PP134840.1	309,286	40.9	complete_genome	Bangladesh	2023
PP134841.1	309,286	40.9	complete_genome	Bangladesh	2021
PP134842.1	309,286	40.9	complete_genome	Bangladesh	2021
PP145080.1	305,118	40.4	complete_genome	United States	2020
PP145081.1	305,075	39.3	complete_genome	United States	2020
PP145082.1	305,076	39.2	complete_genome	United States	2020
PP145083.1	305,122	40.5	complete_genome	United States	2020
PP145084.1	305,119	39.5	complete_genome	United States	2020
PP330954.1	599	49.1	partial_sequence	Egypt	2023
PP663183.1	780	40.5	partial_sequence	India	2022
PP663184.1	417	41.2	partial_sequence	India	2021
PP663185.1	774	40.6	partial_sequence	India	2023
PP816934.1	300	46.3	partial_sequence	China	2025
PP858044.1	786	40.7	partial_sequence	India	2024
PQ206394.1	5,214	40.0	partial_sequence	Thailand	2023
PQ206395.1	1,808	44.5	partial_sequence	Thailand	2023
PQ206396.1	1,795	43.5	partial_sequence	Thailand	2023
PQ206397.1	1,593	44.6	partial_sequence	Thailand	2023
PQ206398.1	1,209	44.8	partial_sequence	Thailand	2023
PQ206399.1	1,150	42.8	partial_sequence	Thailand	2023
PQ206400.1	1,097	38.8	partial_sequence	Thailand	2023
PQ206401.1	1,051	41.8	partial_sequence	Thailand	2023
PQ281795.1	776	40.6	partial_sequence	India	2020
PQ281796.1	524	47.7	partial_sequence	India	2010
PQ379869.1	541	48.6	partial_sequence	India	2024
PQ824250.1	281,051	41.0	complete_genome	United States	2023

ACCESSION	LENGTH	GC	COMPLETENESS	COUNTRY	YEAR
PU788204	627	N/A	N/A	N/A	N/A
PU788222	366	N/A	N/A	N/A	N/A
PU788223	516	N/A	N/A	N/A	N/A
PV030922.1	2,045	38.6	unverified	India	2024
PV574461.1	290,653	40.9	complete_genome	Thailand	2022
PX112034.1	289,826	41.1	unknown	Thailand	2022
PX206431.1	277,683	41.2	complete_genome	Thailand	2022
PX273475.1	877	46.4	unverified	N/A	2025
PX273476.1	866	46.1	unverified	N/A	2025
PX273477.1	868	46.5	unverified	N/A	2025
PX273478.1	878	46.4	unverified	N/A	2025
PX273479.1	895	46.6	unverified	N/A	2025
PX591678.1	288,027	41.0	complete_genome	Japan	2024
PX591679.1	288,412	41.1	complete_genome	Japan	2024
PX591680.1	284,746	41.0	complete_genome	Japan	2024
PX591681.1	288,025	41.0	complete_genome	Japan	2024
PX591682.1	288,025	41.0	complete_genome	Japan	2024
PX591683.1	288,409	41.1	complete_genome	Japan	2024
PX591684.1	288,414	41.1	complete_genome	Japan	2024
PX591685.1	288,414	41.1	complete_genome	Japan	2024
PX591686.1	288,367	41.1	complete_genome	Japan	2024
PX591687.1	288,415	41.1	complete_genome	Japan	2024
PX620399.1	296,256	41.0	complete_genome	Thailand	2022
PX667114.1	17,609	41.3	unverified	N/A	2025
PX667115.1	2,722	39.4	unverified	N/A	2025
PX667116.1	3,918	44.8	unverified	N/A	2025
PX667117.1	11,188	44.2	unverified	N/A	2025
PX667118.1	5,133	42.5	unverified	N/A	2025

Sequence

Find motif Find

528 bp, GC 49.6%

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  61  GAAAACCTCC GCATTCCTGT GACTGCTGAG GTTGGATCAG GCTACTTCAA GATGACTGAT
 121  GTGTCCTTTG ACAGCGACAC CTTGGGCAAA ATCAAGATCC GCAATGGAAA GTCTGATGCA
 181  CAGATGAAGG AAGAAGATGC GGATCTTGTC ATCACTCCCG TGGAGGGCCG AGCACTCGAA
 241  GTGACTGTGG GGCAGAATCT CACCTTTGAG GGAACATTCA AGGTGTGGAA CAACACATCA
 301  AGAAAGATCA ACATCACTGG TATGCAGATG GTGCCAAAGA TTAACCCATC AAAGGCCTTT
 361  GTCGGTAGCT CCAACACCTC CTCCTTCACC CCCGTCTCTA TTGATGAGGA TGAAGTTGGC
 421  ACCTTTGTGT GTGGTACCAC CTTTGGCGCA CCAATTGCAG CTACCGCCGG TGGAAATCTT
 481  TTCGACATGT ACGTGCACGT CACCTACTCT GGCACTGAGA CCGAGTAA
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